

SISA Machine Unlearning

Anuj Lalla (B22AI061)
Indian Institute of Technology Jodhpur

Abstract

We implement SISA (Sharded, Isolated, Sliced, Aggregated) training to honor deletion requests without full retraining. Using CIFAR-10 and a small CNN ($\sim 288k$ params), we compare a baseline model against SISA-unlearned models under 1% and 5% deletions and a naive “full retrain after delete” control. We quantify test accuracy, wall-clock savings, and a confidence-based membership-inference (MIA) AUC sanity check. With $K=20$ shards and $S=3$ slices, SISA retrains only the impacted slices (here $15/60=25\%$), preserving utility (max $\Delta\text{accuracy} \approx -0.7$ pp) while saving 75–83% of time vs. full retraining. MIA AUCs remain near chance ($\approx 0.45\text{--}0.47$).

Index Terms

Machine Unlearning, SISA, Selective Retraining, Membership Inference, CIFAR-10, LoRA, Adapters

I. INTRODUCTION

A practical unlearning system should remove the influence of specific data without incurring the cost of full retraining. SISA organizes data into disjoint *shards*, each processed in chronological *slices*; shard models are isolated during training and aggregated at inference. Deletions trigger retraining only from the earliest affected slice within impacted shards, reducing compute. Our draft results (accuracy vs. deletion on p. 2; time saved and MIA on p. 3) preview these trade-offs.

II. METHOD

A. SISA Design

We split the training set into $K=20$ shards and each shard into $S=3$ chronological slices. Training proceeds slice-by-slice *within* a shard, carrying weights forward; across shards, models are isolated. Aggregation uses logit averaging across shard heads.

B. Deletion Handling

Given a deletion set, we mark any shard containing a deleted point and retrain that shard *from the earliest affected slice onward*. Total retrain work scales with (i) the number of impacted shards and (ii) the earliest affected slice per shard.

C. Metrics

We report: (i) Test Accuracy (%), (ii) Wall-clock time (s) and Time Saved (%), and (iii) a confidence-based MIA AUC (higher $\Rightarrow$ stronger membership signal).

III. EXPERIMENTAL SETUP

Dataset/Model. CIFAR-10, small CNN ($\approx 288k$ params). **Budget.** ≤ 20 epochs per slice (we use 15). **SISA Defaults.** $K=20$, $S=3$. **Aggregation.** Logit averaging. **Deletion Scenarios.** 1% and 5%; a `cluster_shards` strategy concentrates deletions into five shards, keeping retrained slices at $5 \times S = 15$.

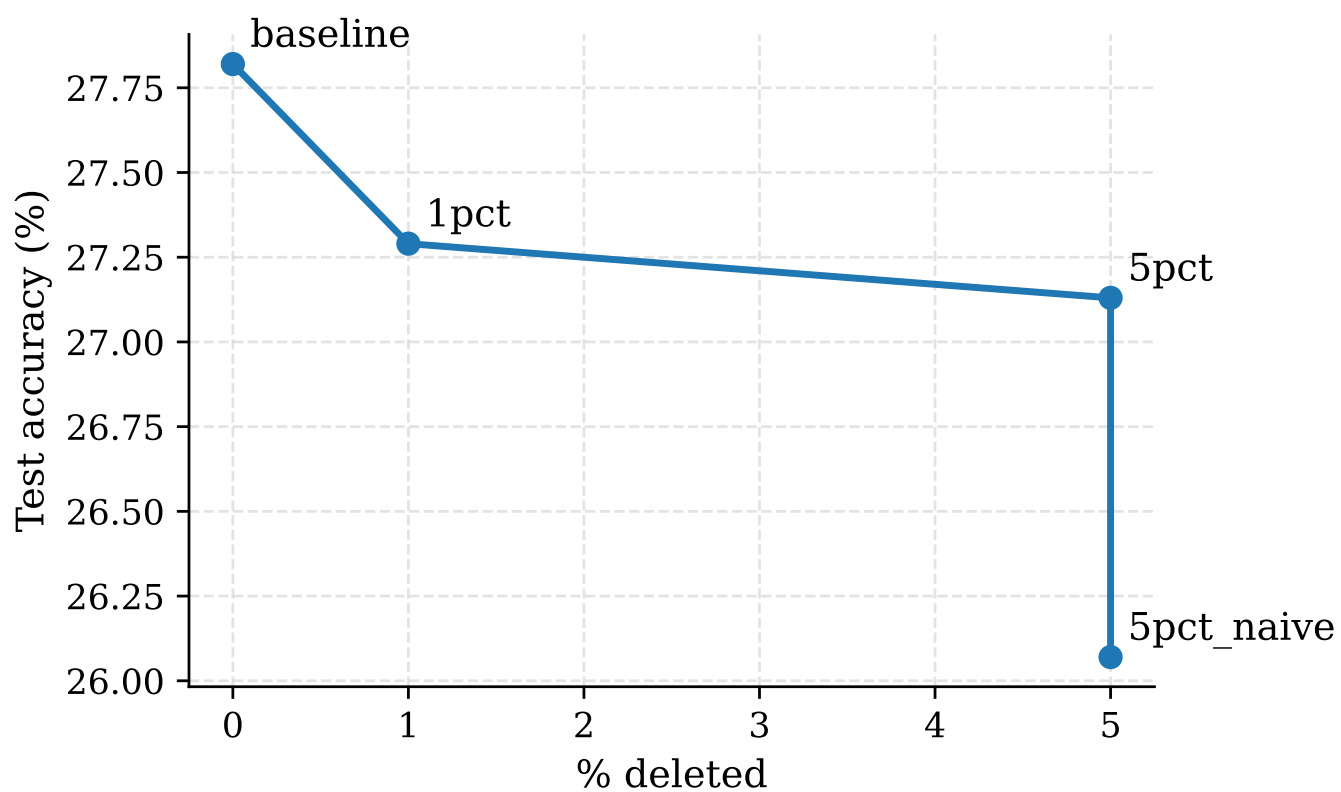


Fig. 1. Test accuracy vs. deletion rate.

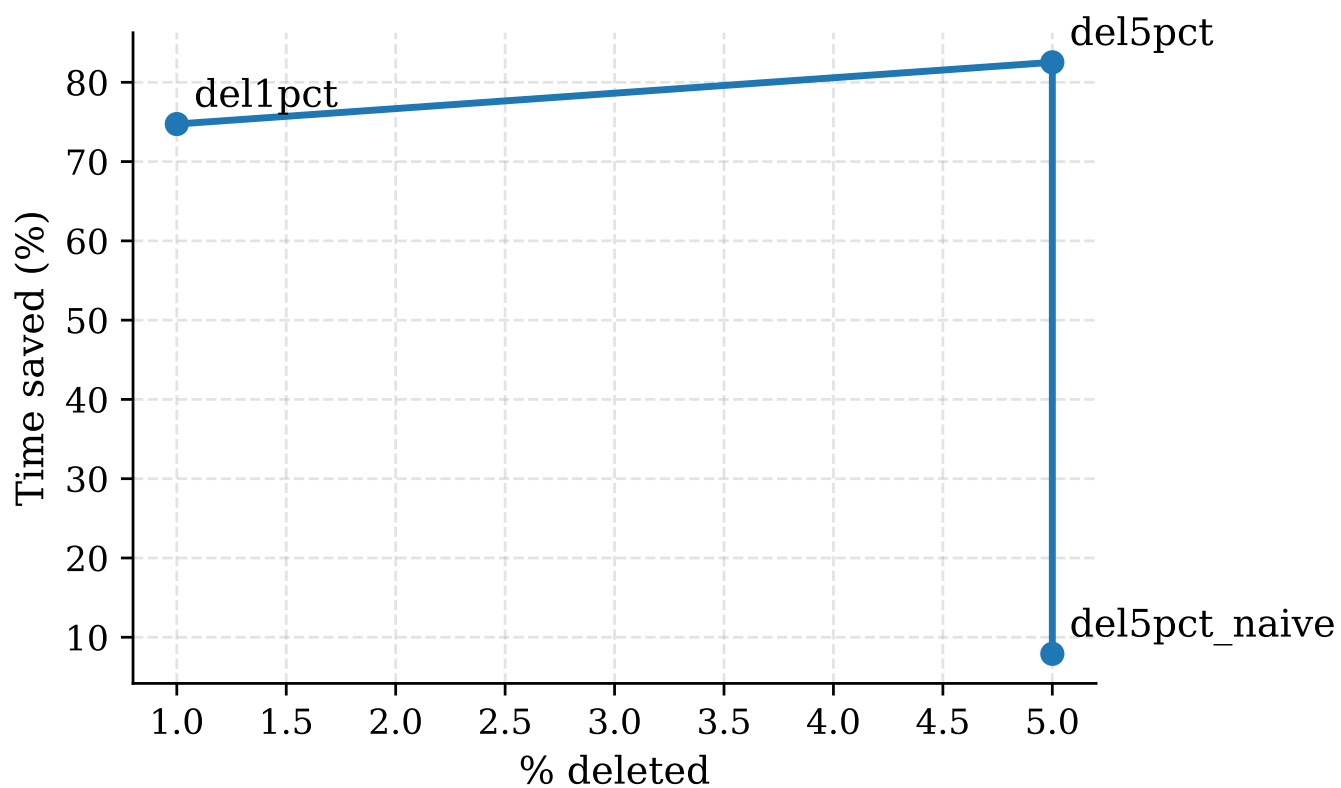


Fig. 2. Time saved vs. deletion rate (relative to baseline).

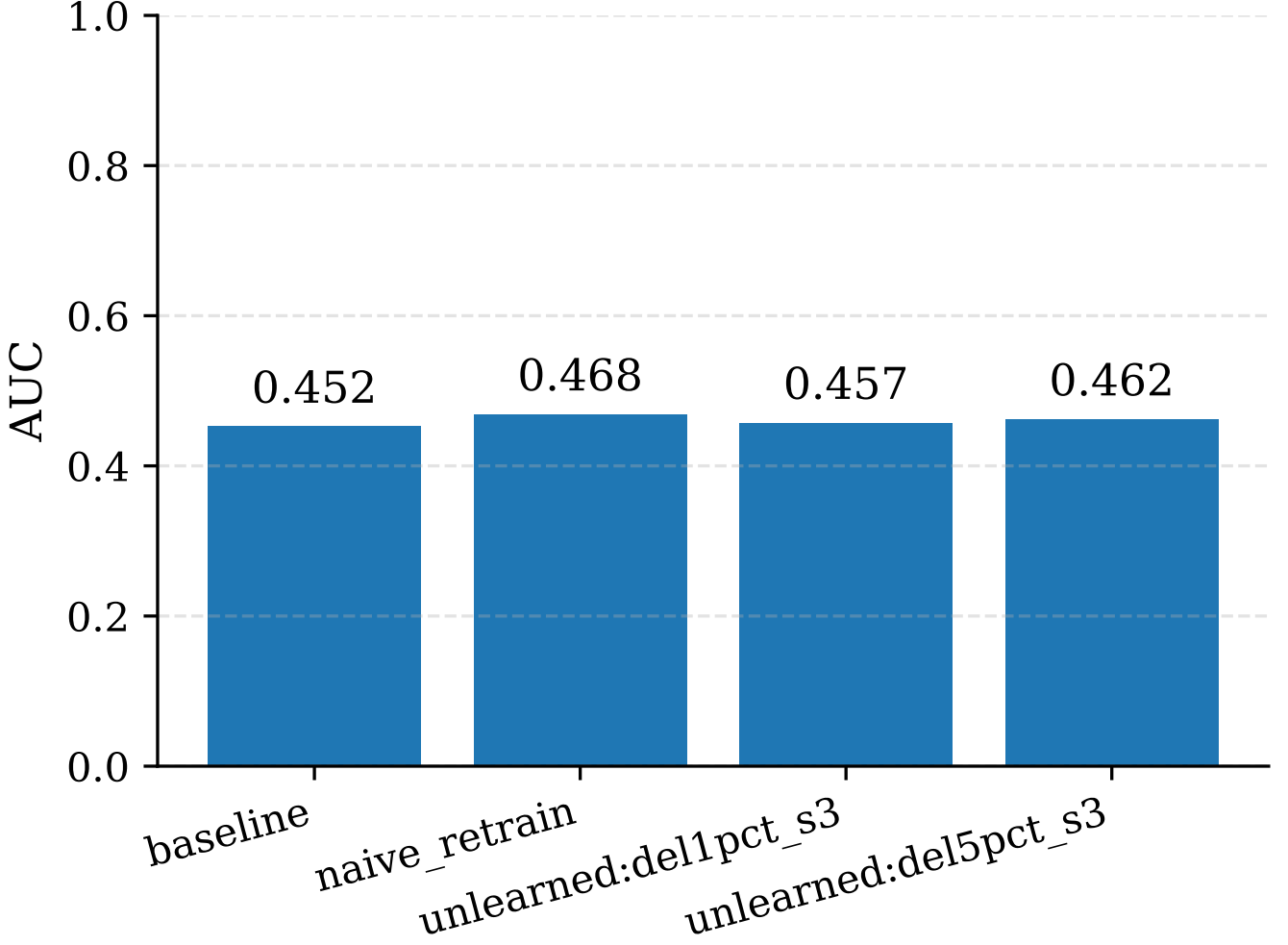


Fig. 3. Membership Inference AUC (confidence attack).

IV. RESULTS

Utility. Baseline accuracy is 27.82%; SISA-1% is 27.29% ($\Delta \approx -0.53$ pp); SISA-5% is 27.13% ($\Delta \approx -0.69$ pp); naive full retrain (5%) is 26.07%. See Fig. 1. (Accuracy plot previewed on page 2 of the draft.)

Efficiency. Baseline wall time is 159.57s. SISA retrains 15/60=25% slices and saves 74.74% (1%) and 82.52% (5%) of time. See Fig. 2. (Time-saved curve previewed on page 3 of the draft.)

Privacy (MIA AUC). Near-chance AUCs: baseline 0.452, naive full retrain 0.468, SISA-1% 0.457, SISA-5% 0.462. See Fig. 3. (Bar chart previewed on page 3 of the draft.)

A. Layout and Timing Summary

Table I summarizes K×S layout and timing (see also the table on p.4 of the draft).

V. TIE-IN TO ADAPTERS/LoRA

SISA’s shard/slice isolation maps cleanly to adapter/LoRA fine-tunes: keep a frozen backbone, store per-shard adapters, and retrain only the earliest affected adapter slices; aggregate via logit averaging or a small gating head. This localizes updates (efficiency) while minimally perturbing global utility (see also the note on p.4 of the draft).

TABLE I
K×S LAYOUT, RETRAINED SLICES, TIMES.

Exp	K	S	Retr./Tot.	Frac.	Time (s)	Saved	Del. %
baseline	20	3	60/60	1.00	159.57	0%	0.0
SISA-1%	20	3	15/60	0.25	40.31	74.74%	1.0
SISA-5%	20	3	15/60	0.25	27.89	82.52%	5.0
naive-5%	20	3	60/60	1.00	146.95	7.91%	5.0

VI. LIMITATIONS & FUTURE WORK

Absolute accuracy is modest by design (small model, per-slice budget). Stronger MIA variants and a uniform deletion ablation are natural next steps.

VII. CONCLUSION

Under $(K=20, S=3)$, SISA satisfies deletions by retraining only impacted slices, preserving utility while saving 75–83% of compute.

REFERENCES

- [1] N. Bourtole *et al.*, “Machine Unlearning,” in *IEEE Symposium on Security and Privacy*, 2021.